

Pavan Kalyan Gollapalli

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience designing and building scalable ETL pipelines, real-time streaming workflows, and large-scale data migration processes. Skilled in PySpark, Spark SQL, Python (Pandas), and FastAPI, with deep expertise in cloud-based data tools and distributed processing. Strong focus on performance optimization, data integrity, and delivering business-ready data solutions across enterprise-scale projects.

SKILLS

Programming & Scripting: PySpark, SQL, Python, Shell Scripting

Big Data & Distributed Processing: Apache Spark, Databricks

Workflow Orchestration: Apache Airflow

ETL & Data Engineering: ETL Development, Data Transformation, CDC, Data Integration, Data Modelling, Data Warehousing

API Development: FastAPI

Cloud & Containerization: AWS (S3, MWAA), Kubernetes, Docker

Version Control & Collaboration: Git, Jira, Confluence

Additional Tools & Technologies: Pandas, JSON Handling, Apache Iceberg, Tableau, Minio

PROFESSIONAL EXPERIENCE

Data Engineer II | Deloitte – Hyderabad | 10/2025 – Present

Project: Realtime Backlog Engine

- Built an end-to-end streaming ETL pipeline using PySpark and Python (Pandas) to ingest and process high-velocity nested JSON data from enterprise order management systems, handling 500K+ records/day with sub-minute latency.
- Designed a multi-layer JSON flattening engine to parse 5–8 levels of nesting across 150+ fields per record, converting raw streaming data into structured, analytics-ready datasets with 99.7% schema consistency.
- Developed and deployed a FastAPI-based data transfer service with REST endpoints to expose enriched order data, cutting average data retrieval time by 40% and enabling self-serve access for downstream teams.
- Tuned PySpark partitioning and parallelism to boost streaming pipeline throughput by 35%, maintaining 99.9% uptime during peak enterprise workloads.
- Implemented error handling, and data validation checkpoints across the pipeline, reducing data loss to near zero and ensuring end-to-end data integrity from ingestion to API delivery.

Data Engineer | Cognizant – Hyderabad | 05/2025 – 10/2025

Project: Payment Data Integration & Processing (Banking Systems)

- Developed and optimized PySpark pipelines for high-volume datasets ingested from AWS S3, improving ETL throughput and accuracy across critical banking data flows.
- Executed data validation and migration checks ensuring consistency and integrity across multiple environments including staging, UAT, and production.
- Automated workflows using Python and CI/CD tooling (Git, automated testing, deployments), enhancing pipeline reliability and reducing manual intervention effort significantly.

Data Engineer | TATA CONSULTANCY SERVICES – Hyderabad | 01/2022 – 05/2025

Project: Data Mart Creation for Geospatial Data

- Designed 2 fact tables and 11-dimension tables for geospatial data marts, enabling efficient analytical queries across large-scale spatial datasets.
- Built automated PySpark/Spark SQL ETL pipelines processing 500 GB–1 TB weekly, with CDC mechanisms ensuring 99.8% update accuracy across incremental loads.
- Optimized workflows via Shell scripting and Spark-submit configurations, reducing manual effort by 12 hours/week and boosting overall processing speed by 40%.

Project: Social Media Data Migration

- Processed 1M+ nested JSON API records daily, flattening 200+ GB of raw data into structured tables — achieving 20% storage savings and 35% faster query performance.
- Built PySpark ETL pipelines handling 500 GB of daily loads with CDC tracking of 200K+ incremental changes, ensuring full data consistency across environments.
- Automated workflows via Shell scripting and Spark-submit, cutting pipeline runtime by 25% and achieving 99.9% uptime across production runs.

Project: Data Privacy

- Migrated 20+ TB of sensitive data across 50+ applications (3,000+ tables) using Teradata, AWS S3, MinIO, and CSV-based transfer pipelines.
- Built and optimized ETL pipelines handling 10 TB of data, boosting pipeline performance by 45% and reducing data errors by 95%, while ensuring full regulatory privacy compliance.

CERTIFICATIONS

- AWS Certified Data Engineer – Associate | Amazon Web Services
- Databricks Certified Data Engineer – Associate | Udemy
- AWS Certified AI Practitioner | Amazon Web Services
- Data Warehouse – The Ultimate Guide | Udemy

EDUCATION

Bachelor of Engineering – Mechanical Engineering | Raghu Engineering College | 2020

ACHIEVEMENTS

- Received the **Excellence Award** at Deloitte for outstanding contribution to the Realtime Backlog Engine project — recognized for delivering a high-performance streaming ETL pipeline with an integrated FastAPI data layer that significantly improved enterprise order data accessibility and end-user experience.
- Created a robust automation framework for QA data validation at TCS, achieving 80%+ reduction in manual processes and accelerating critical issue identification — recognized with accolades for efficiency at **TCS Innovista**.